Bansilal Ramnath Agarwal Charitable Trust's

Vishwakarma Institute of Technology, Pune-37



Department of Computer Engineering

Class and Division	CS-C
Batch Number	03
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Assignment Number	07
Assignment Title	SQL queries
CO Mapped	C07

Theory:

1. A view in SQL is a virtual table created using a SELECT statement. It displays data from one or more base tables without normally storing a separate copy of the underlying data.
2. Views help simplify complex queries, restrict access to specific columns or rows, and provide a customized representation of database information.

Types of views

Simple view: Created from a single table without joins or aggregate functions.
Join view: Created using multiple tables connected through a join condition.

Entities:

1. Department

dept_id (Primary Key)

dept_name

Location

2. Employee

emp_id (Primary Key)

emp_name

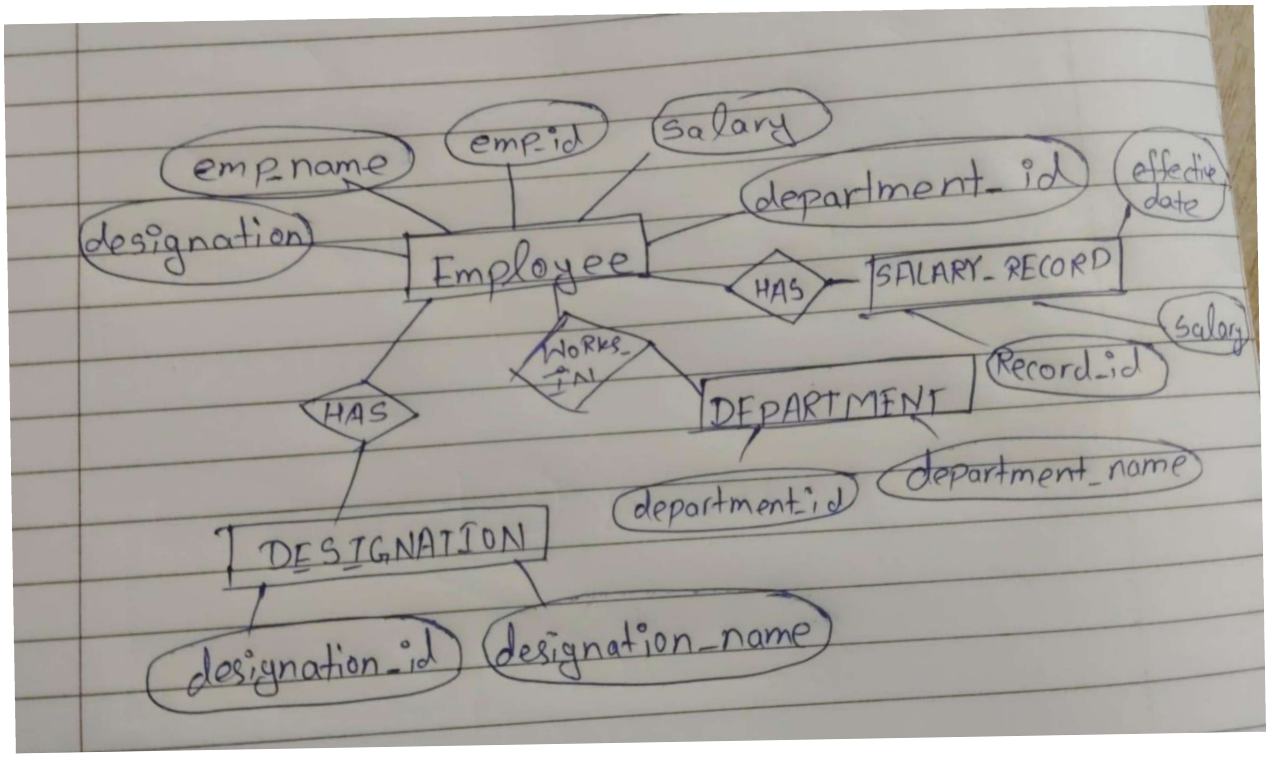
salary

dept_id (Foreign Key)

Relationships:

- Department 1 : N Employee — One department can have multiple employees, but each employee belongs to one department.
- The relationship between Department and Employee is established using dept_id.
- The relationship is implemented through a foreign key constraint in the Employee table

ER Diagram:



Constraints Used:

- Primary Key
- NOT NULL
- FOREIGN KEY
- CHECK

Input:

```
MySQL localhost:33060+ ssl employee4 SQL > SELECT * FROM Employee;
+--------+----------+----------+--------------------+------------+
| emp_id | emp_name | salary   | designation        | department |
+--------+----------+----------+--------------------+------------+
|    101 | Rohit    | 45000.00 | Software Engineer | IT         |
|    102 | Amit     | 38000.00 | Tester             | IT         |
|    103 | Sneha    | 55000.00 | Manager            | HR         |
|    104 | Priya    | 42000.00 | Analyst            | Finance    |
|    105 | Aditya   | 60000.00 | Senior Developer   | IT         |
+--------+----------+----------+--------------------+------------+
5 rows in set (0.0014 sec)
```

Output:

Q1 simple view on a single table with specific conditions.

```
5 rows in set (0.0013 sec)
MySQL localhost:33060+ ssl viewdb SQL > CREATE VIEW HighSalaryView AS
                                     -> SELECT emp_id, emp_name, salary, dept_id
                                     -> FROM Employee
                                     -> WHERE salary >= 30000;
Query OK, 0 rows affected (0.0086 sec)
MySQL localhost:33060+ ssl viewdb SQL > SELECT * FROM HighSalaryView;
+--------+----------+----------+---------+
| emp_id | emp_name | salary   | dept_id |
+--------+----------+----------+---------+
|    101 | Rohit    | 45000.00 |       1 |
|    103 | Sneha    | 55000.00 |       2 |
|    104 | Priya    | 35000.00 |       3 |
+--------+----------+----------+---------+
3 rows in set (0.0026 sec)
MySQL localhost:33060+ ssl viewdb SQL >
```

Q2. Performing select queries on the created view.

```
3 rows in set (0.0026 sec)
MySQL localhost:33060+ ssl viewdb SQL > SELECT emp_name, salary
                                     -> FROM HighSalaryView;
+----------+----------+
| emp_name | salary   |
+----------+----------+
| Rohit    | 45000.00 |
| Sneha    | 55000.00 |
| Priya    | 35000.00 |
+----------+----------+
3 rows in set (0.0020 sec)
MySQL localhost:33060+ ssl viewdb SQL >
```

Q3. update the base table using a view.

```
2 rows in set (0.0019 sec)
MySQL localhost:33060+ ssl viewdb SQL > UPDATE HighSalaryView
                                     -> SET salary = 48000
                                     -> WHERE emp_id = 101;
Query OK, 1 row affected (0.0060 sec)

Rows matched: 1  Changed: 1  Warnings: 0
MySQL localhost:33060+ ssl viewdb SQL > SELECT * FROM HighSalaryView;
+--------+----------+----------+---------+
| emp_id | emp_name | salary   | dept_id |
+--------+----------+----------+---------+
|    101 | Rohit    | 48000.00 |       1 |
|    103 | Sneha    | 55000.00 |       2 |
|    104 | Priya    | 35000.00 |       3 |
+--------+----------+----------+---------+
3 rows in set (0.0021 sec)
MySQL localhost:33060+ ssl viewdb SQL > SELECT * FROM Employee
                                     -> WHERE emp_id = 101;
+--------+----------+----------+---------+
| emp_id | emp_name | salary   | dept_id |
+--------+----------+----------+---------+
|    101 | Rohit    | 48000.00 |       1 |
+--------+----------+----------+---------+
1 row in set (0.0016 sec)
MySQL localhost:33060+ ssl viewdb SQL >
```

Q4 . Creating a view from multiple tables using joins.

```
MySQL localhost:33060+ ssl viewdb SQL > CREATE VIEW EmployeeDepartmentView AS
                                      -> SELECT
                                      ->     E.emp_id,
                                      ->     E.emp_name,
                                      ->     E.salary,
                                      ->     D.dept_name,
                                      ->     D.Location
                                      -> FROM Employee E
                                      -> INNER JOIN Department D
                                      ->     ON E.dept_id = D.dept_id;
Query OK, 0 rows affected (0.0075 sec)
MySQL localhost:33060+ ssl viewdb SQL > SELECT * FROM EmployeeDepartmentView;
+--------+----------+----------+------------+----------+
| emp_id | emp_name | salary   | dept_name  | Location |
+--------+----------+----------+------------+----------+
|    101 | Rohit    | 48000.00 | Computer   | Pune     |
|    102 | Amit     | 25000.00 | Computer   | Pune     |
|    103 | Sneha    | 55000.00 | Mechanical | Mumbai   |
|    105 | Rahul    | 20000.00 | Mechanical | Mumbai   |
|    104 | Priya    | 35000.00 | Electrical | Nashik   |
+--------+----------+----------+------------+----------+
5 rows in set (0.0032 sec)
MySQL localhost:33060+ ssl viewdb SQL >
```

Q5 Studying and demonstrating the restrictions on updatable views by attempting updates on a join-based view.

```
5 rows in set (0.0032 sec)
MySQL localhost:33060+ ssl viewdb SQL > UPDATE EmployeeDepartmentView
                                      -> SET salary = 50000
                                      -> WHERE emp_id = 101;
Query OK, 1 row affected (0.0053 sec)

Rows matched: 1  Changed: 1  Warnings: 0
MySQL localhost:33060+ ssl viewdb SQL > SELECT * FROM EmployeeDepartmentView
                                      -> WHERE emp_id = 101;
+--------+----------+----------+-----------+----------+
| emp_id | emp_name | salary   | dept_name | Location |
+--------+----------+----------+-----------+----------+
|    101 | Rohit    | 50000.00 | Computer  | Pune     |
+--------+----------+----------+-----------+----------+
1 row in set (0.0023 sec)
MySQL localhost:33060+ ssl viewdb SQL >
```

Learning Outcome

Understood the concept and purpose of views in SQL.
Learned how to create a simple view with specific conditions.
Practiced retrieving records from a view using SELECT queries.

Conclusion

In this practical session, I successfully created and used SQL views to retrieve and manage data from the Employee and Department tables. I demonstrated simple views, conditional data retrieval, updates through a view, and views created using joins.